

Executive Summary

Hello! I am Farheena, a pre-final year Electrical and Electronics Engineering student with a strong foundation in core electrical concepts. Skilled in Python, C, MATLAB, AutoCAD and acquired Hands-on experience in power electronics, UPS systems, embedded developments, IoT communication, RTL design, and 3D Printing.

I am actively seeking internships to apply my skills and contribute to real-world engineering challenges.

Skills & Interests

- Programming Languages: Python and C
- Layout Drawing using AutoCAD
- Hardware Description Languages: Verilog & System Verilog
- Circuit Simulations using MATLAB
- PCB designing
- 3D Printing

Soft Skills

- Problem-solving and analytical thinking
- Project and Event Management
- Team Work and Time Management

Education

2023-2027

Prince Shri Venkateshwara Padmavathy Engineering College (Anna University)

• Pursuing B.E.(Electrical and Electronics Engineering), Current CGPA- 8.3

2022-2023

Alwin International Public School, Padappai

• Higher Secondary Education(Affiliated to CBSE) , Secured 74.2%

2020-2021

D.A.V.Public School ,Velachery

• Secondary Education(Affiliated to CBSE) , Secured 83.2%

Projects

• VoltAware – IoT-Based Real-Time Power Line Break Detection System

Designed an intelligent monitoring system for low-voltage AC distribution overhead conductors to detect power line faults in real time. Reduced fault detection time by approximately **3–5 seconds** compared to conventional methods, enabling faster response and improved system reliability.

Tools & Technologies: AutoCAD (circuit design), LTSpice (simulation), 3D Printing (prototype development), IoT Communication

• ReLive – AI & Blockchain-Enabled Smart Swappable LFP Battery System

Developed a sustainable mobility solution for electric vehicles using AI-driven optimization and blockchain integration for battery management. Enabled prediction of battery longevity and degradation using AI models with an accuracy of **~85–90%**.

Tools & Technologies: EasyEDA (PCB design), Proteus (simulation), MATLAB, Visual Studio Code, AI Tools

Technical Internship Experiences

1. Novateur Electrical and Digital Systems Pvt Ltd (Numeric UPS – A Sub-Brand of Legrand)

Duration: June 2025 – July 2025 **Role:** Technical Intern

• Implemented design, testing, and validation of 3-phase UPS systems while analysing power electronics circuits, control strategies, and thermal management. Gained hands-on exposure to industrial power systems, improving understanding of system performance and efficiency

2. Taras Systems and Solutions – Internship Training on VLSI Design and Verification

Duration: June 2025 – December 2025 **Role:** Student Intern

• Designed RTL modules using System Verilog and contributed to FPGA-based development including simulation and debugging. Performed verification using EDA Playground (Aldec Riviera-PRO), strengthening skills in HDL design and validation

Certifications & Accomplishments

1. Completed BS in Data Science and Application -Foundation Level at IIT Madras.

2. Active Member at Tinkerers' Lab of Maker Bhavan Foundation.

3. Organized and hosted college-level hackathons and technical events